STAT210 Statistical Methods I: Linear Model

Lec 08: Multiple Linear Regression - Inference

Zhaoxia Yu
Professor, Department of Statistics

2026-10-19

Section 1

Review

Review: Lecture 01-06

- Lectures 01-06: SLR: least squares estimate, inference (CI, prediction, t-test, F-test), diagnostics, transformations, ANOVA, lack of fit test.
- Lecture 07: Multiple Linear Regression (MLR): least squares estimate.
- This lecture: MLR inference (CI, t-test, linear functions, ANOVA).
- Heads up: next lecture (Lecture 09): Sums of squares, more on F-tests and ANOVA.

Lecture 07:

- MLR: In a multiple linear regression (MLR) model, we have one response variable y and multiple predictor variables $X_1, X_2, \ldots, X_p$.

$$y = \beta_0 + \beta_1 x_1 + \cdots , \beta_p x_p + \epsilon$$

- For individual i, $i = 1, \ldots, n$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$$

MLR Notations and Assumptions

- y_i is the response variable for individual i.
- x_{ij} is the value of predictor variable j for individual i.
- ϵ_i's are errors:
  - The assumptions on ϵ_i's are the same as those in SLR.
  - weaker version: ϵ_i's are independent with mean 0 and constant variance σ^2.
  - Strong version: ϵ_i's are i.i.d. $N(0, \sigma^2)$.
- The parameters to be estimated are $\beta_0, \beta_1, \ldots, \beta_p$ and σ^2.

Matrix Form

- In matrix form, we have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

- More compactly,

$$\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\epsilon}$$

- Assumptions: weaker version: $E(\boldsymbol{\epsilon}) = \mathbf{0}$ and $Var(\boldsymbol{\epsilon}) = \sigma^2\mathbf{I}_n$; strong version: $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$.

Normal Equations: Matrix Form

- For general p, the normal equations can be written as:

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$$

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Fitted values and Residules

- The fitted values can be computed as:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$$

- The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is a projection matrix, i.e.,
  - $\mathbf{H} = \mathbf{H}^T$ (symmetric) and
  - $\mathbf{H}^2 = \mathbf{H}$ (idempotent).
- The residuals can be computed as:

$$\hat{\epsilon} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\beta} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}$$

- $\mathbf{I}_n - \mathbf{H}$ is also a projection matrix.

SSE and MSE

- The sum of squared errors (SSE) can be computed as:

$$\begin{aligned} SSE &= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \hat{\epsilon}^T\hat{\epsilon} = (\mathbf{y} - \mathbf{X}\hat{\beta})^T(\mathbf{y} - \mathbf{X}\hat{\beta}) \\ &= [(\mathbf{I}_n - \mathbf{H})\mathbf{y}]^T(\mathbf{I}_n - \mathbf{H})\mathbf{y} \\ &= \mathbf{y}^T(\mathbf{I}_n - \mathbf{H})\mathbf{y} \end{aligned}$$

- The mean squared error (MSE) is given by:

$$MSE = \frac{SSE}{n - (p+1)}$$

  where $n - (p+1)$ is the degrees of freedom (d.f.) for error.
- An unbiased estimator of σ^2 is given by $s^2 = \hat{\sigma}^2 = MSE$.

Section 2

t-test

Distribution of $\hat{\beta}_j$

- Under the strong version of the model assumptions, i.e., $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, or equivalently, ϵ_i's are i.i.d. $N(0, \sigma^2)$, we have:

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1})$$

- This is because a linear function of a multivariate normal random vector is still a multivariate normal random vector.
- The variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ is given by:

$$Var(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

The Estimated Variance-Covariance Matrix of $\hat{\beta}$

- An estimator of the variance-covariance matrix of $\hat{\beta}$ is given by:

$$\widehat{Var}(\hat{\beta}) = s^2(\mathbf{X}^T\mathbf{X})^{-1}$$

  where $s^2 = MSE$.
- For example, in the height data set, the fitted Model is

$$\hat{c.height} = 16.26 + 0.29 \cdot f.height + 0.46 \cdot m.height$$

- The estimated variance-covariance matrix of $\hat{\beta}$ can be obtained using the `vcov` function in R.

```
height.data <- read.csv("data/height_data.csv")
height.lm= lm(c.height ~ f.height + m.height, data=height.data)
vcov(height.lm)
```

```
##             (Intercept)    f.height    m.height
## (Intercept) 200.4619078 -0.88888954 -2.18030362
## f.height     -0.8888895  0.02274562 -0.01083111
## m.height     -2.1803036 -0.01083111  0.04612353
```

The Estimated Variance-Covariance Matrix of $\hat{\beta}$

- Let's verify the calculation of the estimated variance-covariance matrix of $\hat{\beta}$ using the formula.

```
X <- model.matrix(height.lm)
XtX_inv <- solve(t(X) %*% X)
s2 <- summary(height.lm)$sigma^2
vcov_manual <- s2 * XtX_inv
vcov_manual
```

```
##             (Intercept)    f.height    m.height
## (Intercept) 200.4619078 -0.88888954 -2.18030362
## f.height     -0.8888895  0.02274562 -0.01083111
## m.height     -2.1803036 -0.01083111  0.04612353
```

The Estimated Variance-Covariance Matrix of $\hat{\beta}$

- Based on the var-cov matrix, we can extract the estimated variances and covariances of the regression parameters. For example,
  - the estimated variances are on the diagonal of the matrix.
  - the estimated variance of a linear combination of regression parameters can be computed using the var-cov matrix.

t-test For β_j

- The LSE $\hat{\beta}_j$ follows a normal distribution:

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj})$$

  where $[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$ is the j-th diagonal element of the matrix $(\mathbf{X}^T\mathbf{X})^{-1}$.

- An estimator of the variance of $\hat{\beta}_j$ is given by:

$$\widehat{Var}(\hat{\beta}_j) = s^2[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$$

  where $s^2 = MSE$.

- The standard error (SE) of $\hat{\beta}_j$ is given by:

$$SE(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}}$$

Derive t-test For β_j

- It can be shown that $\hat{\beta}_j$ and s^2 are independent, and

$$\frac{(n-(p+1))s^2}{\sigma^2} = \frac{SSE}{\sigma^2} \sim \chi^2_{n-(p+1)}$$

- Therefore, we have

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}}} \Big/ \sqrt{\frac{(n-(p+1))s^2}{\sigma^2(n-(p+1))}} \sim t_{n-(p+1)}$$

t-test For β_j

- To test the significance of each predictor variable, we can use the t-test:

$$t = \frac{\hat{\beta}_j - \beta_{j0}}{SE(\hat{\beta}_j)}$$

  where β_{j0} is the hypothesized value of β_j under the null hypothesis.
- Under the null hypothesis $H_0 : \beta_j = \beta_{j0}$, the test statistic follows a t-distribution with $n - (p + 1)$ degrees of freedom:

$$t \sim t_{n-(p+1)}$$

Example: t-statistic for β_f

- In the height data set, to test $H_0 : \beta_f = 0$ vs. $H_a : \beta_f \neq 0$, we have

```
beta_f_hat <- coef(height.lm)["f.height"]
se_beta_f <- summary(height.lm)$coefficients["f.height", "Std. Error"]
beta_f_0 <- 0
t_stat <- (beta_f_hat - beta_f_0) / se_beta_f
t_stat
```

```
## f.height
## 1.955048
```

```
#two-sided p-value
p_value <- 2 * pt(-abs(t_stat), df = height.lm$df.residual)
p_value
```

```
##   f.height
## 0.05653888
```

Example: t-statistic for β_f

- Compare the result to the summary output:

```
summary(height.lm)
```

```
##
## Call:
## lm(formula = c.height ~ f.height + m.height, data = height.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.0987 -2.8324 -0.0332  2.2544  7.1131
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.2580    14.1585   1.148   0.2567
## f.height      0.2949     0.1508   1.955   0.0565 .
## m.height      0.4628     0.2148   2.155   0.0363 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.568 on 47 degrees of freedom
## Multiple R-squared:  0.2128, Adjusted R-squared:  0.1793
## F-statistic: 6.352 on 2 and 47 DF,  p-value: 0.003615
```

Example: CI for β_f

- A $(1-\alpha)100\%$ confidence interval for β_j is given by:

$$\hat{\beta}_j \pm t_{\alpha/2,n-(p+1)} \cdot SE(\hat{\beta}_j)$$

where $t_{\alpha/2,n-(p+1)}$ is the $(1-\alpha/2)$ quantile of the t-distribution with $n-(p+1)$ degrees of freedom.

```
alpha <- 0.05
t_alpha_2 <- qt(1 - alpha / 2, df = height.lm$df.residual)
ci_lower <- beta_f_hat - t_alpha_2 * se_beta_f
ci_upper <- beta_f_hat + t_alpha_2 * se_beta_f
ci <- c(ci_lower, ci_upper)
ci
```

```
##     f.height     f.height
## -0.008550141  0.598257182
```

```
# compare to confint function
confint(height.lm, "f.height", level = 1 - alpha)
```

```
##                   2.5 %    97.5 %
## f.height -0.008550141 0.5982572
```

Section 3

A Linear Function of β's

Inference of a linear combination of β's

- The mean response at given predictor values is a linear combination of β's. We have seen that in SLR.
- In the height example, one interesting question is whether the effect of father's height is the same as that of mother's height on child's height, i.e., testing $H_0 : \beta_f = \beta_m$ vs. $H_a : \beta_f \neq \beta_m$.
- This is equivalent to testing $H_0 : \beta_f - \beta_m = 0$ vs. $H_a : \beta_f - \beta_m \neq 0$.
- An obvious estimate of $\beta_f - \beta_m$ is $\hat{\beta}_f - \hat{\beta}_m$.

Variance of A Linear Combination of β's

- It can be "standardized'' to form a t-test statistic if we can compute the standard error of $\hat{\beta}_f - \hat{\beta}_m$.
- The variance of $\hat{\beta}_f - \hat{\beta}_m$ can be computed using the var-cov matrix of $\hat{\boldsymbol{\beta}}$. Specifically,

$$\widehat{Var}(\hat{\beta}_f - \hat{\beta}_m) = \widehat{Var}(\hat{\beta}_f) + \widehat{Var}(\hat{\beta}_m) - 2\widehat{Cov}(\hat{\beta}_f, \hat{\beta}_m)$$

```
vcov(height.lm)
```

```
##                (Intercept)     f.height     m.height
## (Intercept) 200.4619078 -0.88888954 -2.18030362
## f.height     -0.8888895  0.02274562 -0.01083111
## m.height     -2.1803036 -0.01083111  0.04612353
```

Example: SE for $\beta_f - \beta_m$

- Thus, the estimated variance of $\hat{\beta}_f - \hat{\beta}_m$ is given by:

$$\widehat{Var}(\hat{\beta}_f - \hat{\beta}_m) = 0.02274562 + 0.04612353 - 2 * (-0.01083111) = 0.0905313$$

- The standard error (SE) of $\hat{\beta}_f - \hat{\beta}_m$ is given by:

$$SE(\hat{\beta}_f - \hat{\beta}_m) = \sqrt{Var(\hat{\beta}_f - \hat{\beta}_m)} = \sqrt{0.09053137} = 0.3008843$$

Example: t-test for $\beta_f - \beta_m$

- The t-test statistic is given by:

$$t = \frac{(\hat{\beta}_f - \hat{\beta}_m) - 0}{SE(\hat{\beta}_f - \hat{\beta}_m)} = \frac{0.2949 - 0.4628}{0.3008843} = -0.5580218$$

- The p-value can be found using the t-distribution with $n - (p + 1) = n - 3$ degrees of freedom.

Example: CI for $\beta_f - \beta_m$

- A $(1-\alpha)100\%$ confidence interval for $\beta_f - \beta_m$ is given by:

$$\hat{\beta}_f - \hat{\beta}_m \pm t_{\alpha/2,n-(p+1)} \cdot SE(\hat{\beta}_f - \hat{\beta}_m)$$

where $t_{\alpha/2,n-(p+1)}$ is the $(1-\alpha/2)$ quantile of the t-distribution with $n-(p+1)$ degrees of freedom.

$$CI : (0.2949 \pm 2.011741 \cdot 0.3008843) = (-0.31, 1.07)$$

t-test for a linear combination of β's

- Let $L = c_0\beta_0 + c_1\beta_1 + c_2\beta_2 + \cdots + c_p\beta_p$ be a linear combination of regression parameters.
- An estimator of L is given by:

$$\hat{L} = c_0\beta_0 + c_1\hat{\beta}_1 + c_2\hat{\beta}_2 + \cdots + c_p\hat{\beta}_p$$

- The variance of $\hat{L}$ can be computed using the var-cov matrix of $\hat{\beta}$:

$$\widehat{Var}(\hat{L}) = \mathbf{c}^T\,\widehat{Var}(\hat{\beta})\mathbf{c} = s^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}$$

  where $\mathbf{c} = [c_0, c_1, c_2, \ldots, c_p]^T$.
- The standard error (SE) of $\hat{L}$ is given by:

$$SE(\hat{L}) = s\sqrt{\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c}}$$

t-test for a linear combination of β's

- To test $H_0 : L = l_0$ vs. $H_a : L \neq l_0$, we can use the t-test:

$$t = \frac{\hat{L} - l_0}{SE(\hat{L})}$$

  where l_0 is the hypothesized value of L under the null hypothesis.
- Under the null hypothesis $H_0 : L = l_0$, the test statistic follows a t-distribution with $n - (p + 1)$ degrees of freedom:

$$t \sim t_{n-(p+1)}$$

Example: t-test for $\beta_f - \beta_m$

- In the height data set, to test $H_0 : \beta_f - \beta_m = 0$ vs. $H_a : \beta_f - \beta_m \neq 0$, we have

```
beta_f_hat <- coef(height.lm)["f.height"]
beta_m_hat <- coef(height.lm)["m.height"]
se_beta_f <- summary(height.lm)$coefficients["f.height", "Std. Error"]
se_beta_m <- summary(height.lm)$coefficients["m.height", "Std. Error"]
cov_beta_fm <- vcov(height.lm)["f.height", "m.height"]
beta_diff_hat <- beta_f_hat - beta_m_hat
se_beta_diff <- sqrt(se_beta_f^2 + se_beta_m^2 - 2 * cov_beta_fm)
beta_diff_0 <- 0
t_stat <- (beta_diff_hat - beta_diff_0) / se_beta_diff
t_stat
```

```
##   f.height
## -0.5582448
```

```
#two-sided p-value
p_value <- 2 * pt(-abs(t_stat), df = height.lm$df.residual)
p_value
```

```
## f.height
## 0.579327
```

Section 4

R

R

- There are existing R packages and functions to perform inference for a linear combination of β's:
  - the `confint` function in base R can also handle linear combinations of parameters.
  - the `linearHypothesis` function in the `car` package.
  - the `glht(height.lm, linfct = "f.height - m.height = 0")` in the `multcomp` package produces both CI and t-test results.

Section 5

A Linear Test <–> A Reduced Model

Example: Heights: Full Model

- To find reduced model under $H_0 : \beta_f = \beta_m$, we will re-parameterize the full model.
- The full model is

$$c.height = \beta_0 + \beta_f \cdot f.height + \beta_m \cdot m.height + \epsilon$$

This can be rewritten (re-parameterized) as:

$$\begin{aligned} c.height &= \beta_0 + \beta_f \cdot f.height + \beta_m \cdot m.height + \epsilon \\ &= \beta_0 + \beta_f \cdot f.height + \beta_f \cdot m.height + (\beta_m - \beta_f) \cdot m.height \\ &= \beta_0 + \beta_f(f.height + m.height) + \beta_d \cdot m.height + \epsilon \end{aligned}$$

where $\beta_d = \beta_m - \beta_f$.

Example: Heights: Reparameterized Full Model

```
height.lm.reparam= lm(c.height ~ I(f.height + m.height) + m.height, data=height.data)
summary(height.lm.reparam)
```

```
##
## Call:
## lm(formula = c.height ~ I(f.height + m.height) + m.height, data = height.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.0987 -2.8324 -0.0332  2.2544  7.1131
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              16.2580    14.1585   1.148   0.2567
## I(f.height + m.height)    0.2949     0.1508   1.955   0.0565 .
## m.height                  0.1680     0.3009   0.558   0.5793
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.568 on 47 degrees of freedom
## Multiple R-squared:  0.2128, Adjusted R-squared:  0.1793
## F-statistic: 6.352 on 2 and 47 DF,  p-value: 0.003615
```

Example: Heights: Reduced Model

- The re-parameterized ***full*** model is:

$$c.height = \beta_0 + \beta_f(f.height + m.height) + \beta_d \cdot m.height + \epsilon$$

  where $\beta_d = \beta_m - \beta_f$.
- Under $H_0 : \beta_f = \beta_m$, we have $\beta_d = 0$. Thus, the ***reduced*** model is:

$$c.height = \beta_0 + \beta_f(f.height + m.height) + \epsilon$$

Example: Heights: Reduced Model in R

```
height.data <- read.csv("data/height_data.csv")
height.lm.reduced= lm(c.height ~ I(f.height + m.height), data=height.data)
summary(height.lm.reduced)
```

```
##
## Call:
## lm(formula = c.height ~ I(f.height + m.height), data = height.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.3015 -2.8019  0.0952  2.3054  7.1330
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              18.6541    13.3951   1.393 0.170158
## I(f.height + m.height)    0.3571     0.1007   3.546 0.000885 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.542 on 48 degrees of freedom
## Multiple R-squared:  0.2076, Adjusted R-squared:  0.1911
## F-statistic: 12.57 on 1 and 48 DF,  p-value: 0.000885
```

Example: Heights: Reduced Model in R

- We can also compare the two models using ANOVA

```
height.lm.full= lm(c.height ~ f.height + m.height, data=height.data)
height.lm.reduced= lm(c.height ~ I(f.height + m.height), data=height.data)
anova(height.lm.reduced, height.lm.full)
```

```
## Analysis of Variance Table
##
## Model 1: c.height ~ I(f.height + m.height)
## Model 2: c.height ~ f.height + m.height
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     48 602.35
## 2     47 598.39  1    3.9677 0.3116 0.5793
```

Section 6

ANOVA Table for MLR

ANOVA Table for MLR

- The ANOVA table for MLR is similar to that for SLR, with the following differences:
  - The d.f. for regression is p, the number of predictor variables.
  - The d.f. for error is $n-(p+1)$, where n is the number of observations.
  - The total d.f. is still $n-1$.
- The F-statistic is given by:

$$F = \frac{MSR}{MSE} = \frac{SSR/p}{SSE/(n-(p+1))}$$

The Full Model

- The ***full*** model is:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$$

- The alternative hypothesis for the overall F-test in MLR is:

$$H_a : \text{at least one } \beta_j \neq 0, j = 1, 2, \ldots, p$$

- We denote the sum of squared errors (SSE) for the full model as SSE.

The Reduced Model

- The null hypothesis for the overall F-test in MLR is:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

- The ***reduced*** model under the null hypothesis is:

$$y_i = \beta_0 + \epsilon_i$$

- The sum of squares errors (SSE) for the reduced model is denoted as $\sum_{i=1}^{n}(y_i - \bar{y})^2$, which is the SST.

Derive the F-statistic for MLR

- Similar to SLR, we define $SSR = SST - SSE$.
- The F-statistic tests the following hypotheses:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

vs

$$H_a : \text{at least one } \beta_j \neq 0, j = 1, 2, \ldots, p$$

Derive the F-statistic for MLR

- Under the strong version of the model assumptions, we have:

$$\frac{SSR/p}{\sigma^2} \overset{H_0:}{\sim} \chi^2_p$$

and

$$\frac{SSE/\sigma^2}{n-(p+1)} \sim \chi^2_{n-(p+1)}$$

- Since SSR and SSE are independent, we have

$$F = \frac{(SSR/p)/\sigma^2}{(SSE/(n-(p+1)))/\sigma^2} \sim F_{p,n-(p+1)}$$

Example: Heights: ANOVA Table

```
height.lm.reduced= lm(c.height ~ 1, data=height.data)
anova(height.lm, height.lm.reduced)
```

```
## Analysis of Variance Table
##
## Model 1: c.height ~ f.height + m.height
## Model 2: c.height ~ 1
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1     47 598.39
## 2     49 760.14 -2   -161.75 6.3524 0.003615 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```